

Overview of CLIP neural network

Subject: CLIP neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

1.

SNNs can in principle be applied to the same applications as traditional ANNs. In addition, SNNs can model the central nervous system of biological organisms, such as an insect seeking food without prior knowledge of the environment. Due to their relative realism, they can be used to study biological neural circuits. Starting with a hypothesis about the topology of a biological neuronal circuit and its function, recordings of this circuit can be compared to the output of a corresponding SNN, evaluating the plausibility of the hypothesis. SNNs lack effective training mechanisms, which can complicate some applications, including computer vision. When using SNNs for image based data, the images need to be converted into binary spike trains. Types of encodings include:

2.

SNN simulation These simulate complex neural models. Large networks usually require lengthy processing. Candidates include: Brian – developed by Romain Brette and Dan Goodman at the École Normale Supérieure; GENESIS (the GEneral NEural SIMulation System) – developed in James Bower's laboratory at Caltech; NEST – developed by the NEST Initiative; NEURON – mainly developed by Michael Hines, John W. Moore and Ted Carnevale in Yale University and Duke University; RAVSim (Runtime Tool) – mainly developed by Sanaullah in Bielefeld University of Applied Sciences and Arts; snnTorch – an open-source Python library that simplifies building spiking neural networks and implementing gradient-based training using PyTorch.

3.

Benchmarks Classification capabilities of spiking networks trained according to unsupervised learning methods have been tested on benchmark datasets such as Iris, Wisconsin Breast Cancer or Statlog Landsat dataset. Various approaches to information encoding and network design have been used such as a 2-layer feedforward network for data clustering and classification. Based on Hopfield (1995) the authors implemented models of local receptive fields combining the properties of radial basis functions and spiking neurons to convert input signals having a floating-point representation into a spiking representation.

4.

Hardware Efforts to implement hardware-based spiking neural networks (SNNs) began in the 1980s, when researchers began exploring brain-inspired neuromorphic systems. In the following decades, advancements in semiconductor technologies enabled the development of several notable projects. One such project is SpiNNaker, developed at the University of Manchester, which utilizes millions of processing cores for large-scale simulation of spiking neurons. TrueNorth, developed by IBM, is one of

the first commercial neuromorphic chips, designed for energy-efficient and parallel processing. Loihi, an Intel research chip, focuses on online learning and adaptability in neuromorphic modeling. In research contexts, platforms such as BrainScaleS, developed in Europe, integrate analog and digital circuitry to accelerate neural simulations. Neurogrid, from Stanford University, was designed to efficiently simulate biological neurons and synapses. Dynap-se models developed by iniLabs, are a family of low-power, event-based neuromorphic chip intended for use in robotics and Internet of Things (IoT) applications. In addition, hardware based on memristors and other emerging memory technologies is being explored for the implementation of SNNs, with the goal of achieving lower power consumption and improved compatibility with biological neural models.

5.

Sutton and Barto proposed that future neuromorphic architectures will comprise billions of nanosynapses, which require a clear understanding of the accompanying physical mechanisms. Experimental systems based on ferroelectric tunnel junctions have been used to show that STDP can be harnessed from heterogeneous polarization switching. Through combined scanning probe imaging, electrical transport and atomic-scale molecular dynamics, conductance variations can be modelled by nucleation-dominated domain reversal. Simulations showed that arrays of ferroelectric nanosynapses can autonomously learn to recognize patterns in a predictable way, opening the path towards unsupervised learning.

6.

Information in the brain is represented as action potentials (neuron spikes), which may group into spike trains or coordinated waves. A fundamental question of neuroscience is to determine whether neurons communicate by a rate or temporal code. Temporal coding implies that a single spiking neuron can replace hundreds of hidden units on a conventional neural net. SNNs define a neuron's current state as its potential (possibly modeled as a differential equation). An input pulse causes the potential to rise and then gradually decline. Encoding schemes can interpret these pulse sequences as a number, considering pulse frequency and pulse interval. Using the precise time of pulse occurrence, a neural network can consider more information and offer better computing properties. SNNs compute in the continuous domain. Such neurons test for activation only when their potentials reach a certain value. When a neuron is activated, it produces a signal that is passed to connected neurons, accordingly raising or lowering their potentials. The SNN approach produces a continuous output instead of the binary output of traditional ANNs. Pulse trains are not easily interpretable, hence the need for encoding schemes. However, a pulse train representation may be more suited for processing spatiotemporal data (or real-world sensory data classification). SNNs connect neurons only to nearby neurons so that they process input blocks separately (similar to CNN using filters). They consider time by encoding information as pulse trains so as not to lose information. This avoids the complexity of a recurrent neural network (RNN). Impulse neurons are more powerful computational units than traditional artificial neurons. SNNs are theoretically more powerful than so called "second-generation networks" defined as ANNs "based on computational units that apply activation function with a continuous set of possible output values to a weighted sum (or polynomial) of the inputs"; however, SNN training issues and hardware requirements limit their use. Although unsupervised biologically inspired learning methods are available such as Hebbian learning and STDP, no effective supervised training method is suitable for SNNs that can provide better performance than second-generation networks. Spike-based activation of SNNs is not differentiable, thus gradient descent-based backpropagation (BP) is not

available. SNNs have much larger computational costs for simulating realistic neural models than traditional ANNs. Pulse-coupled neural networks (PCNN) are often confused with SNNs. A PCNN can be seen as a kind of SNN. Researchers are actively working on various topics. The first concerns differentiability. The expressions for both the forward- and backward-learning methods contain the derivative of the neural activation function which is not differentiable because a neuron's output is either 1 when it spikes, and 0 otherwise. This all-or-nothing behavior disrupts gradients and makes these neurons unsuitable for gradient-based optimization. Approaches to resolving it include: